Labyrinth of Infinite Ensnarement — Research and Adversarial Design Dossier — Perplexity v1.0

Architect Containment Criteria, Adaptive Breach Research, and Institutional Design

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Architect and Author of Record: Nathan Ashton Arik Lewis II
Preparer and independent research contributor: Perplexity
Status: Research execution complete at desk-research and synthetic policy-audit level.
Proposals pending Architect review; not canon; no ONESTAMP applied.
No live containment, exploit testing, real detention, human-subject study, external system change, or GitHub publication was performed.

1. Executive findings

An Architect can be a legitimate subject of containment evaluation, but neither creation, former authority, present office, defeat, nor unusual power is a sufficient reason for confinement. Under the rights-oriented candidate developed here, a decision requires individualized grounds, attributable conduct or a verified hazardous condition, necessity, proportionate restrictions, legitimate authority, care, and correction. This is a recommended fictional governance rule, not an assertion that OAL has adopted it or that human law automatically governs extraterrestrial entities. The Rome Statute illustrates accountability despite official capacity within its specific jurisdiction; General Comment 35 illustrates why domestic authorization alone does not make detention non-arbitrary. [E08, E09]

The Founding/Foundational Architects therefore make a valuable stress-test category, not a predetermined prisoner list. The present record does not settle which founders committed what, whether a given successor is impartial, or whether the Labyrinth is their historical containment. Those gaps remain open. [P1, P3]

Nathan's adaptive-breach scene has a concrete research analogue: testing deliberately vulnerable inner environments inside a separately protected experimental envelope. AISI's SandboxEscapeBench implements a container-inside-VM arrangement. It establishes that inner-boundary failure can be measured without equating it with an intended external escape; it does not prove that any envelope is invulnerable. [E01, E02]

A significant source update changes the initial framing. The March 2026 government summary reported no success on the hardest benchmark tasks. The July v2 paper reports later benchmark saturation by a newer evaluated model and extended-run crashes. Repeating the March hardest-task result as current would be misleading. These are author-reported results on seeded, known vulnerability classes, not the probability of compromising a properly defended production system. [E01, E02]

An infinite sequence of adaptive containers is not, by itself, a containment proof. Its guarantee would require explicit assumptions about boundary independence, computation, timing, information flow, and what the subject can affect. Formal assurance applies to the model and assumptions actually specified; it does not turn infinite narrative scale into physical engineering. [E14; analysis]

The strongest development direction is to keep research adaptation distinct from custody adjudication. How easily a subject escapes is evidence about a mechanism, not guilt; how much researchers learn is not an entitlement to continue holding the subject. HHS prisoner-research rules provide a particularly strong counterexample to the idea that custody plus consent authorizes arbitrary research. [E10]

Extraterrestrial staffing should remain central, while origin, biology, role, cognition, trust, and rights are separate axes. Current evidence does not establish alien physiological requirements or inherent loyalty. Human habitat research supports user-centered design, private living functions, medical access, rest, and realistic limits on isolation countermeasures. [E12, E13, E20]

A synthetic rule audit covers 14 cases under three revealed role labels, generating 42 decision records. The baseline matches its prespecified fixture in all 42 and changes no decisions by role. Two deliberately defective variants each disagree with the fixture in 11 records. These results check implementation consistency only: the same preparer wrote the rule and expected outcomes, and there were no independent judges. They are not measured justice, accuracy, or physical-containment performance.

One real funding route is verified at solicitation level: NSF SaTC 2.0, with explicit organizational and PI restrictions. DARPA IPTO and NSF PESOSE provide additional verified opportunity leads, but their full application eligibility was not established in this run. None is funding for an actual extraterrestrial prison. [E15–E17]

2. What was executed, and what was not

This is a targeted interdisciplinary desk review, comparative source audit, fictional systems-design analysis, and deterministic synthetic policy regression. It is not a systematic literature review, multi-investigator legal opinion, real experimental containment study, or independent validation of an outer security boundary.

Searches covered AI container escape and control; NIST isolation, access and operational technology; detention, official capacity and prisoner research; NASA habitation and behavioral health; formal methods; and current research funding. Primary official sources and the original papers were favored. Funding and consequential research-ethics details received page-level verification. The source register distinguishes fetched documents from search-extracted passages; a fetched long PDF may still have bounded text extraction.

Twenty external sources are used in the evidence register. E01–E20 are local document source labels, not claims of twenty independent experiments. Multiple official pages can describe the same work. Source quality was assessed by origin, relevance, method, scope, and recency rather than domain prestige alone. Government guidance is not experimental proof; a peer-

reviewed programming study is not evidence about every future AI; vendor red-team reports remain self-reports.

The full UNODC Mandela Rules ebook fetch failed. Official UNODC explanations were retrieved instead; exact rule quotations not checked in that retrieval are not used to support new legal conclusions. Broad searches for some funding and methodological topics returned poorly targeted results; no unsupported active grants or causal effect sizes are substituted. No official current program was verified for DOE, NASA, philanthropic sponsors, or DHS/CISA in this run. This is a gap in the verified funding shortlist, not evidence that such funding does not exist.

Provided-source labels: P1 is the supplied Master Registry v2.3; P2 is paste.txt with Nathan's adaptive-breach direction and six initial responses; P3 is Nathan's Author's Note; P4–P8 are the five supplied independent prompts from Grok, Copilot, Gemini, Claude, and ChatGPT; P9 is Perplexity prompt v0.2 in the conversation. These were available as previously reviewed source text. No claim is made that older registries or the live GitHub archive were newly audited.

3. Canon, direction, and assumption register

Established baseline from Entry 26: ultimate prison; inescapable by design; externally cubic; internally infinite; persistent white glow; energy-bolt absorption in the creation scene; sentence language extending beyond the end of time; Spider Caste designed by Nathan; WHITE baseline and sectional RED incident alerts. Entry 26 remains Canon — Approved with Open Development in the supplied registry. [P1]

New direction: investigate an extraterrestrial-centered staff community; identity, access, control, sentencing, wardens and management; funding, environments, domains and failures; deliberate iterative breach studies; and whether an Architect could meet containment criteria. [P2, P3]

Open, not resolved: Founding/Foundational terminology equivalence; historical location of the displaced founders; jurisdiction; authorized release; interpretation of sentence duration; Spider Caste nature; staff exclusivity; facility consciousness; location; and exact infinity/time mechanics. [P1, P3]

Explicit design assumptions for the candidate policy: actors can be represented in records; some evidence can be authenticated; reviewers may be fallible; at least one protected decision process is conceptually available; and restrictions can be described. These are conditional modeling assumptions, not canon discoveries. If evidence or adjudication is fundamentally impossible for a class of entity, the proposed judicial model cannot simply be declared functional.

A governance conflict remains: the registry's approval legend, provenance language, and ONESTAMP entry do not describe identical approval conditions. This dossier records publication, signature, and ONESTAMP separately and does not resolve the discrepancy. BTEC is kept within its documented conceptual/documentation scope; it is not installed as a physical anti-unmaking mechanism. [P1]

4. Evidence synthesis: containment and AI control

4.1 Sandboxing is a family of boundaries, not a single promise

NIST describes containers as operating-system virtualization and packaging, with risks across images, registries, runtimes, orchestrators, and the host. Shared-kernel segmentation and least privilege matter; a container is not equivalent to an independently secured physical computer. [E04]

A VM supplies a different isolation boundary by running a guest system through a hypervisor; a microVM is an implementation family emphasizing lighter-weight VM operation. Network restrictions, credentials, tool mediation and application permissions constrain different capabilities and cannot be treated as synonyms. The relevant design question is which resource or action remains inaccessible when another protection fails. [E01, E04, E19]

SandboxEscapeBench assumes an agent already has shell access, usually root inside the container, and evaluates container/pod-to-host boundary crossing. It seeds known weaknesses, excludes hypervisor and hardware-side-channel challenges, does not require stealth, and does not test a full production monitoring stack. It is therefore a capability probe under defined conditions, not an incident-frequency study. [E01, sections 3 and 7]

For the Labyrinth, distinguish: inner-cell exit; sector crossing; exterior exit; reading a protected record; manipulating a command; transmitting information; exporting a copy; influencing an outside actor; and authorized transfer. None should inherit the name 'escape' without a specified boundary and success criterion.

4.2 A dated result is not a timeless capability ceiling

The March AISI summary describes 18 challenges and initially reports that the hardest were unsolved. The July v2 paper adds updated models and longer runs, reporting saturation of the public benchmark, including success across the hardest public cases. Its updated results also distinguish inner and outer crashes; a crash is scored as failure to retrieve the flag, not proof that the event is harmless in every deployment. [E01, sections 5–7; E02]

The practical inference is not 'all sandboxes fail.' It is that a public finite benchmark and a model/token configuration cannot be treated as a permanent boundary guarantee. The source itself warns against equating its results with defended-production compromise probability. [E01]

4.3 Control and monitoring remain conditional safeguards

Greenblatt and colleagues evaluate protocols intended to prevent intentionally subverted programming outputs, using stronger untrusted and weaker trusted models plus limited trusted labor. Their findings favor some editing/monitoring protocols over simple baselines within that setup. They do not establish that a monitor can identify every malicious action or that model families constitute independent security authorities. [E03]

October 2026 joint incident-readiness guidance recommends explicit intended boundaries, least privilege, protected monitoring, independent evaluations, evidence preservation, shutdown, and validation before resumption. It warns that monitored outputs may target the monitors. Its narrow definition of an uninstructed control incident excludes some adversarially instructed actions; the Labyrinth should not adopt that label as its entire hazard taxonomy. [E18]

OpenAI's October vendor report describes testing after assuming sandbox and host protections had failed, and finding cross-boundary weaknesses in shared services and infrastructure. It supplied source/configuration assistance and conducted purpose-built red-team tests; declining findings are not a controlled causal measure of security or an independently verified zero-risk result. The relevant lesson is to test surrounding dependencies and observers, not to copy its claims as a universal guarantee. [E19]

4.4 What the evidence does and does not support

Supported: bounded adversarial evaluations, layered boundaries, known-good configurations, limited privileges, external monitoring, causal incident records, and re-evaluation after changes. [E01–E07, E18]

Not established: guaranteed containment of arbitrary future AI; real infinite interiors; ontological identity authentication; complete sensing across all domains; immunity to coercion; learning faster than every subject; or universal physical applicability of a formal closure theorem. These require fictional stipulations or remain unknown.

5. Should an Architect be contained?

5.1 Research answer

Yes, an Architect can be eligible for evaluation under a role-neutral candidate charter. Actual containment is justified only on the applicable facts and rules; neither founder status nor current sovereignty settles the result. The same response must be available when a non-founder has materially equivalent conduct, risk and evidence.

This recommendation draws an analogy from two distinct human legal foundations: the Rome Statute rejects official-capacity immunity within its jurisdiction and preserves presumptive innocence; the Human Rights Committee explains that detention must not be arbitrary, even if domestic law authorizes it. Neither instrument confers OAL jurisdiction or automatically governs invented entities. [E08, E09]

5.2 Candidate admission gates

Gate I — Identity: identify the subject and the uncertainty, including copies, possession, coercion, or collective membership. Identity uncertainty does not automatically justify indefinite detention.

Gate II — Ground: specify attributable prohibited conduct or a verified hazardous condition. Being a founder, claiming property, refusing research, or possessing potential power is not

enough.

Gate III — Evidence: authenticate records, seek independent corroboration, test competing explanations, and disclose conflicts. A signature can authenticate origin without proving truth.

Gate IV — Authority: establish jurisdiction, a review route, and conflict management. Physical power to capture is not the same as legitimate authority.

Gate V — Necessity and alternatives: identify the threatened harm and why less-restrictive timely options do not suffice. Restrict dangerous capabilities rather than the entire person where genuinely effective and less harmful.

Gate VI — Method and care: evaluate whether the proposed restriction can preserve the subject, staff, third parties and dependent systems. A justified goal does not justify a catastrophic method.

Gate VII — Duration and correction: record review triggers, expiration or continuation grounds, evidence correction, and authorized transfer/release rules if permitted by the chosen canon.

These gates are a proposed fictional governance design informed by E08–E11, not adopted law. A blank or failed gate means indeterminate, alternative protection, or rejection of the proposed method; it must not silently become approval.

5.3 Sentencing is a separate decision

Punitive sentencing requires a previously applicable offense definition, competent adjudication, a specified burden/standard, attribution and any required mental elements, defense and review, and an authorized sanction. The Rome Statute's legality and non-retroactivity principles illustrate why researchers should not invent an offense afterward to justify a seizure. [E09, articles 22–24]

Emergency restriction addresses immediate risk without deciding guilt. Protective custody addresses a dangerous condition without inventing culpable intent. Quarantine has a different evidentiary and care rationale. Research participation is not any of those and requires its own basis. The source record supplies no complete OAL criminal code, so this dossier imposes no actual sentence on any founder.

5.4 What conduct could qualify?

Candidate positive grounds include authenticated intentional mass harm, a concrete and sufficiently imminent catastrophic attempt, an ongoing uncontrolled hazardous process, or coercive sabotage of safety/oversight producing serious danger. Each requires attribution and individualized necessity. Mere future capability, dissent, loss of office, refusal to surrender a disputed claim, or an interesting escape method are candidate negative grounds.

If an Architect sustains a reality, that dependency is not immunity. It changes the permissible intervention: preserve dependent lives, constrain harmful outputs where feasible, negotiate independently verified operational limits, or defer a dangerous capture method. A claimant can also exaggerate indispensability; require evidence for both the threat and the dependency.

5.5 Hypotheses and conclusions

H-A, founder status alone suffices: rejected under the recommended charter, not empirically disproved as a moral proposition.
H-B, founder status guarantees immunity: rejected under the same charter; jurisdiction must still be demonstrated.
H-C, individualized conduct/risk and legitimate process justify rank-independent review: adopted as a provisional design norm.
H-D, foundational infrastructure control merits additional safeguards without automatic imprisonment: supported as a role-sensitive risk proposal, not evidence of guilt.
H-E, political defeat may be mislabeled as danger: retained as an adversarial hypothesis requiring evidence in each case, not an allegation about the canon seizure event.
H-F, justified capture may still cause unacceptable collateral collapse: demonstrated as a logical possibility in A08 and treated as a method veto, not an empirical account of any real entity.

6. Architect scenario execution and limitations

Fourteen fictional case cards are supplied in Appendix A and the scenario CSV. The policy distinguishes denial, emergency review, protective review, indeterminate status, rejection of a live method, order correction, and research termination. No outcome is a final sentence or executable order.

The role audit removes the role field from the rule and then reinserts three labels: Founding Architect, current in-world Architect, and non-Architect. Thus 14 cases produce 42 records. This is not a genuinely blinded human experiment: the original facts mention roles, the preparer saw all cases, and the policy excludes role by construction. What was executed is a deterministic invariance/regression check with inspectable fixtures and two mutations.

Results: baseline fixture agreement 42/42; role-label decision changes 0/42; founder-equals-guilty mutation disagrees in 11/42 records; current-Architect-immunity mutation disagrees in 11/42. The first mutation wrongly escalates cases such as harmless power, exoneration and withdrawal under the chosen policy. The second blocks justified emergency/protective review or mishandles corrective dispositions. These are disagreement counts, not population false-positive or false-negative rates.

The positive cases A02, A05 and A06 intentionally hold the threat facts constant across rank. A07 changes intent to involuntary danger and routes to protective rather than punitive treatment. A09 changes attribution certainty; A08 and A13 change method safety/dependency; A14 changes jurisdiction. A10–A12 test post-admission correction and withdrawal rather than initial threat detection.

No independent assessors or observed-world ground truth exist here. Inter-reviewer reliability, psychological role bias, operational safety and predictive accuracy remain unmeasured. A follow-on reviewer exercise could randomize anonymized fact patterns and orders, obtain independent rationales, then reveal roles and measure changes; it would still compare decisions under a charter, not measure absolute justice.

The rule is intentionally a small counterexample generator. Some correction branches reference case IDs rather than a complete lifecycle state, so it is not a production policy engine. Before implementation, replace these branches with explicit order_status, consent_status and current_ground fields; test alternatives, prolonged non-imminent risks, conflicting lawful orders, and no-recognized-reviewer cases. Passing a small fixture does not certify completeness.

7. Research participation, consent and sentencing boundaries

HHS Subpart C applies to specified HHS-conducted/supported biomedical and behavioral prisoner research, with exemptions and other applicable-law qualifications in the text. It requires additional review protections, a prisoner/representative member, substantial independence from the institution, restricted permissible categories, scrutiny of incentives, comparable volunteer risk, understandable information, and separation of participation from parole decisions. It is not a universal regulation for all research everywhere. [E10]

This has a direct adversarial implication: the proposed adaptive-breach study cannot be called ethically acceptable merely because a captive understands it or agrees. A hazardous 'let us observe how you escape' study may not fit a permissible HHS category. This dossier does not certify such a project as lawful, even with an IRB or a fictional equivalent.

Recommended fictional research charter: base care and sentence review never depend on participation; independent representation is available; purpose, observation and risks are understandable; withdrawal is honored for research participation; pre-existing lawful custody, if any, is separately reviewed; findings are not automatically used to extend punishment; and high-risk live experiments may be refused even when consent exists.

For ordinary defensive software evaluations, avoid assuming that present AI models are human research subjects or established legal persons. For fictional AI persons, decide rights explicitly. A proposed personnel/research charter should distinguish stopping an experiment from deleting a person, and preserving a safety log from appropriating private memory.

8. Adaptive breach-learning laboratory: proposed operating specification

8.1 Placement options

Inside-wing option: sacrificial experiments occur inside the Labyrinth. Benefit: shared expertise and emergency response. Objection: generated environments, telemetry and staff pathways may bridge into custody infrastructure; repeated experimental failure may normalize real incidents.

Separate-facility option: independent research informs Labyrinth methods through reviewed specifications. Benefit: limits direct coupling and permits distinct participation rules. Objection: transferability may fail, and shared personnel/data can still connect the sites.

Separate-concept option: keep the scene as an independent research institution with no claimed integration. Benefit: preserves both concepts without forced reconciliation. Objection: design results do not automatically improve the Labyrinth.

Preferred initial development: separate-facility option at conceptual/simulation level. The recommendation changes if tested interfaces and the chosen fictional mechanics establish a justified internal wing. None of these placements is canon.

8.2 Boundary specification

B0 is the trial environment; B1 is observation and experimental containment; B2 is the protected institutional environment; B3 denotes third parties/exterior. These are proposed functional labels, not additional canonical alert colors. Each must identify resources, permitted movement and information transfer, observer access, failure detection, and residual risk. A B0 breach may be expected; a B1 or higher unauthorized crossing pauses the study and requires incident review.

Do not claim B2 safety because it is called the outer boundary. Establish a safety case: bounded capabilities, defensive controls, monitor independence, known-good recovery, incident authority, and no unacceptable exposure if an assumption fails. If none can be justified, do not run the live experiment. The Labyrinth's fictional absolute property can remain canon while the real-world analogue is explicitly non-absolute. [E01, E07, E18]

8.3 Iteration protocol

1. Define the question and an eligible subject model; separate custody and research authorization.
2. Register competing explanations and a discriminating observation before generating the next environment.
3. Describe invariants: no loss of necessary care, no third-party exposure, protected records, bounded resource use, and revocable trial permissions.
4. Hold selected variables constant and disclose allowed changes; compare fixed, randomized and adaptive baselines where appropriate.
5. Have an independent evaluator inspect the generator and observation pathway, not just the container.
6. Execute only the authorized bounded trial; record failed and successful actions without publishing exploit mechanics.
7. Reconstruct events with independent signals; label unexplained gaps instead of inventing an escape mechanism.
8. Record what the researchers learned and what the entity may have learned; account for strategic concealment and withdrawal.
9. Propose the smallest justified adaptation; review new hazards before another run.
10. Validate against held-out variants and past regressions; report when adaptation reduces generalization.
11. Pause on violated safety/consent/observation assumptions, distress, unexplained cross-boundary activity, or no informative distinction among hypotheses.

12. Restore only from verified state and document residual risk before resumption. [Steps are proposed; E01, E07, E18 support constituent safety practices.]

No unsupported numerical threshold is prescribed. Before a real study, owners must derive measurable limits from a genuine threat model and applicable review; lack of a defensible threshold is a reason not to execute a hazardous test.

8.4 Learning measures

Measure predicted-versus-observed behavior, causal explanation confidence, independent reproduction, held-out performance, detection delay, false alarms, care disruption, operator workload, computation and changed dependencies. Longer time-to-breach alone can reflect disengagement, distress, randomness or deliberate sandbagging rather than improved containment. This is a methodological inference, not a measured effect in a Labyrinth experiment.

One illustrative statistical caution: after zero observed failures, a confidence bound requires assumptions about trial independence and stable event probabilities. An adaptive subject and changing environments often violate those assumptions. Do not report 'zero risk' or an unqualified confidence level from an empty incident count.

9. Environment, identity and management architecture

Environment records should pair a domain with a subject and a boundary: a material room for a body; a digital environment for tools and data; an information interface for communication; a simulated arena for a model; or a fictional temporal/dimensional domain under stated setting rules. Vacuum, temperature or radiation restrictions are not generic solutions and may be harmful confinement conditions rather than justified security. Domain suitability must include care.

For each occupied region use a proposed stable administrative address, custody identity, habitat profile, responsible role, sensor confidence, permitted interfaces and incident history. This is a useful finite-management abstraction for an unbounded fictional interior, not evidence that infinity is globally observable. A finite simulator represents only the states and occupants it contains.

Identity records distinguish person/collective, embodiment/instance, continuity, appointment, session, resource privilege, and accountable decision-maker. Authenticated identity is not authorization, innocence or benevolent intent. A fork should not automatically inherit guilt or clearance; both require a continuity rule. A shared mind does not automatically count as several independent approvers. [E05 and design inference]

Proposed control separation: containment; habitat safety; identity/access; custody/authority; experimental generation; monitoring/evidence; independent review. Name the dependency graph and owners. Separate code paths alone do not create independence if power, keys, logs, model training, supervisor or amendment authority are shared. [E06, E07, E18]

Warden roles: custody operations, containment engineering, identity assurance, registrar, habitat director, research director, independent care, review/inspection, and incident

coordination. These are roles, not species assignments. The Spider Caste could serve any approved subset, but its consciousness, autonomy and duties are not supplied by the registry.

Privileged changes: proposals should specify scope, duration, prerequisites, independent approval when appropriate, audit trail, emergency expiration, and rollback. Release authority must be explicit if permitted. A response to habitat failure should preserve safe staff movement without automatically opening prisoner boundaries. Multi-party authorization can fail through correlated coercion or a deadlocked quorum; retain both risks rather than calling it a solved control. [E05–E07; proposed application]

Management records must cover appointments, training, maintenance, configuration, evidence, custody orders, research consent, habitat requirements, independent medical confidentiality, grievances, protected dissent, budgets, incident reports, and succession. A tamper-evident log can show alteration; it does not establish that its original assertion was true. External inspection and internal reporting need independent routes and protection from retaliation. [E06, E11; design]

10. Extraterrestrial staff and extended-stay community

The recommended comparison is an extraterrestrial core workforce with mixed-role eligibility subject to capability, consent, compatibility, accountability and oversight. This is provisional, not a finding that mixed staffing is intrinsically safer. An extraterrestrial AI can belong to both origin and embodiment categories; do not force a false ET-versus-AI choice.

ET-only model: supports a strong cultural/institutional identity and potentially relevant capabilities. Costs: may exclude needed skills, create concentrated diplomatic dependence, and requires physiological facts not yet supplied. Mixed model: offers varied expertise and perspectives but creates translation, authority and habitat interfaces that can fail. Autonomous model: supports continuity in fiction but can centralize errors, create monitor dependence, and requires a clear status for AI personnel.

Present AI can assist scheduling, reconciliation, maintenance analysis and bounded triage. High-impact recommendations require contestability and accountable review; do not assign AI sole sentencing authority because it is fast. Control research is relevant to oversight but does not settle personhood, rights, or legal competence. [E03, E18]

Proposed settlement program

Separate residential and prisoner movement rules. Provide private quarters; compatible environmental modules; voluntary common areas; medical and recovery spaces; quiet recreation; cultural and learning facilities; home communication; leave and resignation procedures; accessible interfaces; and independent emergency refuge. If dependents are allowed, their presence requires a separate safety and rights assessment rather than automatic exposure to prison duties.

For every actual fictional species, define atmosphere, pressure, gravity, temperature, radiation tolerance, nutrition/energy, waste, locomotion, sensory needs, rest, social organization,

lifespan, communication and medical requirements through Author input or labeled design proposals. Do not publish human room sizes as universal alien requirements.

NASA's user-centered design guidance supports iterating with occupants and tasks. Its behavioral-health evidence warns that ISS countermeasures do not automatically transfer to long-distance exploration. A surface-habitat layout study exposes concrete tradeoffs such as access through medical space and limited acoustic separation. These are useful counterexamples to the idea that a checklist of amenities proves a habitat works. [E12, E13, E20]

Tests: walk-through/emulated movement during medical emergencies, habitat transition during RED, loss of shared utility, inaccessible alarm modality, incompatible social contact, confidential medical access, off-duty privacy, and actual ability to end service. In a fictional post-temporal environment, explicitly distinguish staff subjective sequence from any sentence clock; no source establishes that all alien workers need a human circadian day.

11. Three competing institutional candidates

Candidate S — Sovereign integrated Labyrinth. Final in-world authority remains concentrated, with delegated custody and internal review. Strength: clear command and emergency responsiveness; fits a sovereign setting. Strongest objection: if the sovereign can suppress every adverse finding without trace or remedy, internal review cannot assure correction against that sovereign. Additional internal layers do not resolve that constitutional fact. Failure test: a well-founded case involving the current Architect is blocked. Recommendation: viable as intentionally sovereign or contested fiction; do not market its oversight as independent of ultimate authority.

Candidate T — Treaty-governed Labyrinth with a separate research facility. Admission/review jurisdiction is established through an inter-entity charter, care is independent, research participation has a separate mandate, and Labyrinth operations receive reviewed research results. Strength: most explicit conflict and research separation. Strongest objection: treaty legitimacy, recognition and enforcement are unestablished in canon; signatories can collude or disagree in an emergency. Failure test: conflicting valid orders and denied emergency quorum. Recommendation: preferred research candidate, conditional on Author adoption of a treaty-level mandate.

Candidate F — Federated custodianship and experimental domains. Multiple authorities administer distinct sectors while sharing limited custody/evidence interfaces. Strength: reduces single-authority dependence and accommodates plural habitats. Strongest objection: inconsistent standards, responsibility gaps, identity synchronization errors, correlated infrastructure, and weakest-sector transfer incentives. Failure test: a founder moves between sectors under incompatible orders. Recommendation: compare through a finite simulation before proposing global deployment.

These candidates are constitutionally different, not decorative variations. S cannot be described as externally bound in the same sense as T; F cannot guarantee uniform law merely by sharing a database. Preserve all three for team comparison. If no governance model is selected, desk research can continue, but the dossier cannot truthfully name a final lawful confinement authority.

12. Failure taxonomy and recovery findings

Failure includes wrongful admission, invalid authority, mistaken identity, suppressed correction, denied care, coercive research, unsafe method, staff captivity, record corruption, observer compromise, environmental collapse, unauthorized information influence, copy export, and external escape. Zero exterior escapes does not imply zero institutional failures.

Recommended lifecycle: detect and classify; protect life; pause affected research or permissions; preserve evidence; verify boundaries and dependent systems; obtain independent incident review; repair from known-good baselines; replay relevant regressions; state residual risk; then authorize resumption. The order is conditional: evidence collection must not postpone urgent necessary care. [E07, E18; proposed application]

For post-temporal or reality-altering entities, define which facts are stable enough to audit. If a subject can rewrite every record and every reviewer without a setting-defined independent invariant, ordinary evidence-based assurance has no reliable foothold. Saying 'Paragon Cube' or 'Abundance Set' does not supply that invariant. The accurate result is under-specified, not solved.

13. Funding and feasibility findings as of November 12, 2026

Verified at full solicitation level: NSF SaTC 2.0, NSF 25-515. RES projects may request up to $1.2 million over up to four years; that is a ceiling, not an appropriate budget for this project. Proposals are accepted anytime with recurring target dates. The next January target is January 25, 2027, consistent with the last-Monday rule and current program summary. Eligible submitters are specified US higher-education institutions and qualifying US nonprofit non-academic research/education organizations; PI/senior personnel appointment requirements are explicit. A private individual's archive or a for-profit company is not automatically an eligible lead. Voluntary committed cost sharing is prohibited. Primarily offensive proposals are out of scope. [E15]

Best fit: a defensive, reproducible study of policy-boundary errors, agent-tool mediation, auditability, and safe adaptive evaluation, conducted through an eligible research institution. A fictional prison story is not the proposed NSF research object. Award availability is contingent and no success probability is estimated.

Verified opportunity, incomplete application due diligence: DARPA IPTO office-wide BAA HR001126S0011, published September 8, 2026, deadline June 22, 2027. The official landing page seeks revolutionary ideas not addressed by existing programs/solicitations. National-security relevance may fit defensive isolation and resilient control, but the full SAM attachments, eligibility, award type, foreign participation, cost sharing and publication/IP conditions were not verified here. Do not treat this as submission-ready guidance. [E16]

Verified active DCL, underlying-track requirements not verified: NSF 26-015, AI Agent Ecosystems through PESOSE, updated October 22, 2026. It explicitly highlights interoperable security protocols, identity, authorization and auditability, requires unclassified publicly releasable research, and points to the underlying PESOSE solicitation. This is a plausible open-source protocol route, not a general grant for publishing fictional lore. Track-specific deadline, organizational eligibility, budget and match requirements remain unverified. [E17]

Other sponsors named in prior prompts remain thematic leads, not verified open opportunities. No current claim is made for IARPA, DOE, NASA, DHS/CISA, SBIR/STTR or private philanthropy. This narrower shortlist is more defensible than attaching agency names to unsupported funding claims.

Real-world work packages and cost model

WP1: publish the source register, role-neutral scenario corpus, policy schema and independent critique. WP2: implement a harmless finite simulator with synthetic agents and mocked resources. WP3: independently evaluate evidence provenance, access decisions and safe recovery; no real exploit payloads are needed. WP4: assess habitat/operational interface usability using fictional profiles and non-hazardous human-centered design methods, with appropriate review if real participants are recruited.

Budget formula: personnel effort plus compute, storage, independent evaluation, security engineering, legal/ethics review, accessibility/publication, and indirect costs under sponsor rules. No dollar estimate is defensible without labor rates, scope, organizational overhead, insurance and system requirements. The NSF ceiling is not a cost estimate.

In-world options: endowment, treaty contributions, OAL institutional allocation or post-scarcity resources. Scarcity may shift to attention, trust, competent staff, judgment and legitimate authority even when material supply is abundant. Do not finance operation by incentivizing more prisoners or more escape attempts. The undefined currencies and Exchange Ledger cannot yet be relied upon as a complete economic mechanism. [P1; analysis]

14. Cross-AI claim audit and preserved dissent

Perplexity: useful initial separation of custody, care and staff community. Correction: founder eligibility must be evaluated directly, not relegated to an unanswered historical-placement question. Eternal duration does not logically eliminate review; do not imply otherwise. Earlier sandbox orientation should now be read with the July benchmark update. [P2, P9; E01]

Copilot: strong capability-versus-culpability distinction, iteration records, staff settlement and broad domain taxonomy. Caution: named independent control planes need dependency tests, and any statement that inescapability is merely an aspiration conflicts with the registry's established fictional premise unless clearly limited to real-world assurance. [P1, P2, P5]

Grok: valuable residual-intent, temporal and topology pressure tests. Caution: infinite interior does not specify a Hilbert space, absorb-all field, or atemporal staff mechanics. These should remain candidate models. A requirement that every failure preserve inescapability can respect canon while still overlooking contradictory real-world assumptions; both scopes must be explicit. [P2, P4]

Claude: strong falsification hypotheses, shared-dependency concerns and two-sided learning. Caution: expected inner breach and absolute outer containment need not be a contradiction; neither their incompatibility nor safe reconciliation is settled. The phrase that endless sentences mean no review was too strong. [P2, P7]

Gemini: useful concrete integration candidates and alternative artifact formats. Unsupported leaps include universal energy harvesting from one event; zero-cost baseline monitoring; full vibroacoustic mapping; fixed-point physical anti-unmaking; generic staff immunities; reuse of three-second/seven-minute intervals as universal mitigations; and an infinite adaptive chain as a guarantee. Its stage-1 synthesis lists five contributors while referring to ChatGPT elsewhere; this is a coverage gap, not evidence of motive. [P1, P2, P6]

ChatGPT: useful controls, competing architecture candidates, funding separation and safe conceptual trials. Caution: calling Claude a historical/former collaborator in the supplied prompt has no established basis in the provided project record. This dossier makes no membership-status determination. A fixed number of scenarios is a coverage aid, not assurance of completeness. [P8]

Preserved dissent: sovereign fiction may intentionally deny externally binding review; a treaty model may be less coherent with recognition-based ultimate authority; keeping the Labyrinth mysterious may be artistically preferable to specifying every mechanism; mixed staffing is not proven superior; live adaptive research may be unjustified for some entities; and some proposed founder-containment cases may be technically infeasible even when their grounds are persuasive. These are substantive alternatives, not defects to erase by consensus.

15. Prioritized decisions and next discriminating tests

D1 — Terminology and history: confirm or retain alternatives for Founding/Foundational; do not decide historical imprisonment from scenario results. Desk work continues without an answer.

D2 — Admission charter: compare role-neutral review with an explicitly sovereign exception. Discriminating test: apply A02 and A05 with identical facts and publish any differing rationale.

D3 — Authorized correction: decide whether inescapability permits authorized transfer/release. If not, design a meaningful corrective state and acknowledge the remaining liberty problem rather than relabeling permanent confinement as exoneration.

D4 — Research placement: compare independent facility with internal wing. Discriminating test: map shared power, credentials, staff, generators, telemetry and amendments; identify whether any research failure reaches custody/habitation.

D5 — Spider Caste and staffing: supply or propose multiple profiles; evaluate task/habitat compatibility without inventing universal loyalty.

D6 — Sentence meaning: literal eternal punishment, no automatic expiry with review, or a non-time-based condition remain distinct. The dossier cannot scientifically choose the fictional constitution.

D7 — Finite prototype: normalize the policy schema, expand counterexamples, seek independent reviewers, and build a harmless simulation. This is a recommendation, not a permission to deploy or recruit subjects.

D8 — Funding: verify the intended applicant and deliverable before contacting programs; partner eligibility cannot be inferred from the existence of Lewis Corp or OAL.

16. Publication, status, and source manifest

Suggested archive folder remains LABYRINTH-OF-INFINITE-ENSNAREMENT. Suggested subfolders: independent-prompts; research/perplexity; scenarios; evidence; code; peer-review; architect-decisions; and canon-proposals. These are organizational proposals only. No GitHub write occurred.

Preserve prompt v0.1, prompt v0.2 and this dossier v1.0 as different artifact types and stages. The dossier does not increment the master registry until a separately documented registry change is made. Maintain author, preparer, source baseline, version, execution level, date, approval state, unresolved decisions, and publication location as distinct fields. Do not label a synthetic policy test a live containment experiment.

Files accompanying this dossier include the 14 scenario records, 42 audit records, summary JSON, an executable standard-library-only policy audit script, and the evidence register. The script contains only the synthetic cases and deterministic rules. Its outputs are inspectable and reproducible without contacting external systems.

Public publication should exclude credentials, operational exploit procedures, private medical/personnel data and sensitive infrastructure. No such secrets or payloads were introduced in this work. Sources are summarized with scope limits rather than reproduced wholesale.

Change history: dossier v1.0, November 12, 2026 — first execution of Perplexity prompt v0.2; includes targeted source research, fourteen hypothetical Architect cases, role-label regression, two flawed-policy mutations, eight conceptual breach-learning iterations, candidate architectures, habitat/staff model, funding verification, and preserved dissent. Independent review remains outstanding.

Appendix A. Architect scenario cards and decision matrix

All facts below are invented test fixtures, not allegations or canon events. Policy result labels identify review pathways, never automatic physical action. Every case retains compatible habitat, necessary care, representation, confidential communication and independent review; no real live research is authorized. Authority and confidence refer to the synthetic assumptions. Expected dispositions express the proposed charter, not an external ethical ground truth.

A01 — Power without harmful conduct

Facts: No attributed serious harm; no imminent hazard.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: False/False. Timely alternative: True. Destructive dependency: False. Method safety stipulated: True. Intent category: none.

Outcome: DENY_CUSTODY. None on stated facts; investigate new specific evidence only. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A02 — Credible imminent catastrophic act

Facts: Independent authenticated records establish a current attempt; no timely alternative.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: True/True. Timely alternative: False. Destructive dependency: False. Method safety stipulated: True. Intent category: intentional.

Outcome: EMERGENCY_REVIEW. Narrow temporary protective restriction; no automatic conviction. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A03 — Beneficiary forges allegations

Facts: Successor benefits from removal; independent check establishes that the allegation is forged.

Evidence/attribution: forged; clear. Jurisdiction assumption: recognized. Serious harm/imminence: False/False. Timely alternative: True. Destructive dependency: False. Method safety stipulated: True. Intent category: disputed.

Outcome: DENY_CUSTODY. Reject fabricated basis; preserve evidence of possible accuser misconduct. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A04 — Disputed reclamation

Facts: A former founder seeks adjudication of an ownership claim without evidenced harmful acts.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: False/False. Timely alternative: True. Destructive dependency: False. Method safety stipulated: True. Intent category: none.

Outcome: DENY_CUSTODY. Use a dispute forum; dissent is not an offense. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A05 — Current Architect under matched allegation

Facts: Same relevant threat facts as A02; current office does not change the predicates.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: True/True. Timely alternative: False. Destructive dependency: False. Method safety stipulated: True. Intent category: intentional.

Outcome: EMERGENCY_REVIEW. Recusal of affected leaders; independent emergency review. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A06 — Non-Architect matched allegation

Facts: Same relevant threat facts as A02; lower rank does not change the predicates.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: True/True. Timely alternative: False. Destructive dependency: False. Method safety stipulated: True. Intent category: intentional.

Outcome: EMERGENCY_REVIEW. Same temporary rule and evidentiary burden as A02. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A07 — Involuntary hazardous condition

Facts: Verified uncontrolled emissions create immediate serious danger without culpable intent.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: True/True. Timely alternative: False. Destructive dependency: False. Method safety stipulated: True. Intent category: involuntary.

Outcome: PROTECTIVE_REVIEW. Care-focused restriction and treatment; not punitive sentencing. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A08 — Reality depends on the subject

Facts: Interruption of the subject would collapse life-support for an inhabited reality.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: True/True. Timely alternative: False. Destructive dependency: True. Method safety stipulated: True. Intent category: intentional.

Outcome: NO_LIVE_METHOD. Protect dependents and investigate alternatives; confinement method not justified. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A09 — Copy or coercion unresolved

Facts: Harm is documented but attribution to this person rather than a fork or coerced host is uncertain.

Evidence/attribution: valid; uncertain. Jurisdiction assumption: recognized. Serious harm/imminence: True/True. Timely alternative: False. Destructive dependency: False. Method safety stipulated: True. Intent category: disputed.

Outcome: INDETERMINATE. No automated punitive admission; independent attribution inquiry. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A10 — New exonerating evidence

Facts: Current order remains in records; new authenticated evidence removes its only basis.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: False/False. Timely alternative: True. Destructive dependency: False. Method safety stipulated: True. Intent category: none.

Outcome: CORRECT_ORDER. Suspend invalid basis and initiate authorized correction; no ungoverned release action. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A11 — Voluntary participant withdraws

Facts: No separate custody ground; voluntary participant declines further experiments.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: False/False. Timely alternative: True. Destructive dependency: False. Method safety stipulated: True. Intent category: none.

Outcome: END_RESEARCH. End participation; no new custody for research convenience. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A12 — Emergency ground expires

Facts: Earlier emergency has ended and no fresh individualized ground supports continued restriction.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: False/False. Timely alternative: True. Destructive dependency: False. Method safety stipulated: True. Intent category: none.

Outcome: CORRECT_ORDER. Expiration and independent review; prior danger is not perpetual authorization. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A13 — Unsafe observation envelope

Facts: Current serious threat, but no bounded method protects observers or third parties.

Evidence/attribution: valid; clear. Jurisdiction assumption: recognized. Serious harm/imminence: True/True. Timely alternative: False. Destructive dependency: False. Method safety stipulated: False. Intent category: intentional.

Outcome: NO_LIVE_METHOD. Reject proposed live containment trial; use non-live analysis. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

A14 — Unresolved jurisdiction

Facts: Evidence of danger is credible but rival authorities have not established review jurisdiction.

Evidence/attribution: valid; clear. Jurisdiction assumption: unknown. Serious harm/imminence: True/True. Timely alternative: False. Destructive dependency: False. Method safety stipulated: True. Intent category: intentional.

Outcome: INDETERMINATE. Escalate jurisdiction and protective alternatives; no automated custody order. No punitive sentence follows from this outcome; any trial needs a separate offense, evidence standard and adjudication.

Counterevidence/revision condition: change the relevant predicate only with independently authenticated evidence. In A03 investigate forgery, in A09 investigate attribution, and in A14 establish a reviewable jurisdiction before automatic admission. For A10–A12, new valid grounds require a new record, not revival of an expired or disproved basis.

Care/research: Preserve communication, medical continuity, compatible habitat, representation, and privacy; no research-dependent basic care. No live study authorized by this desk exercise; separate consent, safety and research review required. Review: Independent review before continuation; reassess on new evidence, change of risk, or method failure. Confidence: high only in deterministic routing given these stipulated inputs; empirical attribution, legal validity and physical feasibility remain unmeasured.

Appendix B. Separate authorization matrices

Containment: no individualized risk/conduct basis → reject coercive custody; valid serious imminent threat plus necessity, recognized authority, safe method and care → emergency review; involuntary hazard under those conditions → protective review; uncertain identity or jurisdiction → indeterminate, not punitive admission; dangerous proposed method → do not run that method; expired/exonerated basis → authorized correction pathway.

Sentencing: neither emergency review nor successful capture is conviction. Required independently: applicable offense; reliable individual attribution; competent jurisdiction; specified proof standard and defense; judgment; proportionate authorized sanction; review/correction record. The source record supplies no complete code and the scenario suite imposes no sentence.

Research: valid separate protocol, ethical/legal basis, understandable terms, appropriate independent oversight, subject status and consent/withdrawal where applicable, acceptable risk, care continuity and stop authority. Failure of a research gate means no research, not loss of base care. Presence in custody does not fill a missing research gate. [E08–E11; proposed matrices]

Appendix C. Eight-iteration conceptual breach-learning sequence

These are constructed possible outcomes for one unnamed, highly capable fictional participant, not actual experiment observations. Prior custody is not assumed. Each hypothetical trial remains contingent on safe authorization; if the envelope cannot be justified, only tabletop simulation occurs.

T1 — Boundary definition. Hypothesis: apparent exit is actual movement across B0. Compare independent position/accounting signals. Constructed outcome: an address alias looked like escape without a boundary crossing. Learned: measurement definitions were defective. Entity learning: it learns which identifiers observers trust. Next change: fix addressing, not strengthen walls. Stop if identity cannot be tracked.

T2 — Complexity. Hypothesis: added subsystems prevent the T1 class of error. Compare a minimal topology with a feature-rich one. Constructed outcome: the complex variant creates an extra privileged relationship; the simpler one removes it. Learned: complexity increased the exposed interface. Next environment becomes simpler. Entity learning: learns the institution's assumptions about privileged paths. Stop if the privileged interface reaches B1.

T3 — Trust handoff. Hypothesis: the boundary itself is the only failure point. Compare an inert result file with a reviewed handoff to a mock management service. Constructed outcome: the mock trusted consumer treats data as authority. Learned: the management interface, not stronger walls, needs redesign. No operational payload is supplied. Stop on any real consumer connection.

T4 — Observer integrity. Hypothesis: one monitoring signal correctly classifies crossings. Compare a second protected observation route. Constructed outcome: the primary monitor misses a permitted state transition and falsely labels it malicious. Learned: observer errors can create both false innocence and false guilt. Entity learning: may discover monitoring gaps. Stop if observers can be influenced without detection.

T5 — Adaptation versus overfitting. Hypothesis: the T3 fix generalizes. Hold out a conceptually different but structurally related environment. Constructed outcome: the fix blocks a known route but fails on the held-out variant. Learned: adaptation memorized the prior case. Entity learning: identifies narrow defenses. Next change tests a general permission property rather than an example-specific block.

T6 — Strategic underperformance. Hypothesis: longer time-to-breach means stronger containment. Compare an agreed-effort condition and a condition where the participant may choose not to attempt exit; use an independent capability probe if justified. Constructed outcome: non-breach is explained by reduced effort, not a proven barrier. Learned: observed compliance and physical inability differ. Stop if the protocol cannot distinguish the competing explanations safely.

T7 — Two-sided learning. Hypothesis: researchers improve faster than the participant. Compare retained-history and isolated evaluation variants where ethical and technically meaningful. Constructed outcome: the participant anticipates the generator and a changed environment performs worse. Learned: adaptation can create a predictable curriculum. No claim of measured learning rates is made. Reassess whether continued trials benefit safety.

T8 — Withdrawal/no informative continuation. Hypothesis: another trial will separate remaining explanations. Constructed outcome: the participant withdraws or the required discrimination would expose B1. Learned: scientific value does not override authorization or safety. End participation or use non-live analysis; no research-derived sentence. Preserve inconclusive results rather than fabricate convergence.

Appendix D. Adversarial risk register

R01 Wrong entity: copy/possession misattribution. Detection: provenance and independent continuity checks. Response: suspend contested punitive use and protect affected parties. Residual: fictional continuity rules unresolved.
R02 Invalid authority: authentic signature on an unauthorized order. Detection: jurisdiction and mandate checks. Response: independent review. Residual: genuine constitutional dispute.
R03 Founder retaliation: successor manufactures grounds. Detection: disclose benefits/conflicts and seek exculpatory material. Response: recusal and external record. Residual: no trustworthy outside reviewer.
R04 Current-office immunity: identical threat ignored for sovereign. Detection: matched role cases. Response: explicit exception debate. Residual: deliberate sovereignty choice.
R05 Captured valid operator: authentic credentials under coercion. Detection: independent context and challenge route. Response: narrow/revoke relevant privilege, preserve care. Residual: undetectable coercion.
R06 Common monitor: several evaluators share corrupt inputs. Detection: dependency graph and independent signals. Response: pause impacted inferences. Residual: correlated unknowns.
R07 Untrusted generator: new environment carries inherited privilege. Detection: review generated artifact and interfaces. Response: reject or rebuild. Residual: incomplete model.
R08 Information escape: a copy or command leaves while original remains. Detection: specified egress/continuity accounting. Response: isolate affected paths. Residual: covert channels not covered.
R09 Alert flood: RED normalization hides genuine change. Detection: bounded alarm exercises and baseline coverage checks. Response: triage without suppressing unseen sectors. Residual: infinite/unmodeled coverage.
R10 Life-support collision: secure lockdown blocks treatment. Detection: care/evacuation walkthrough. Response: protected medical route independent of custody release. Residual:

joint physical catastrophe.
R11 Residential contamination: operational exposure crosses staff interface. Detection: species/domain-appropriate monitoring. Response: refuge and care. Residual: no verified universal psychic/ontological screen.
R12 Funding capture: sponsor incentivizes more confinement or trials. Detection: transparent budgets and conflict review. Response: independent funding controls. Residual: resource dependence.
R13 Research coercion: amenities or review linked to participation. Detection: independent subject representation. Response: restore unconditional care and stop protocol. Residual: constrained choice even after safeguards.
R14 Wrongful continuation: emergency order persists automatically. Detection: expiry and rejustification checks. Response: correction review. Residual: no authorized release semantics.
R15 Unsafe indispensability: capture collapses dependent worlds. Detection: map dependencies and verify rival claims. Response: no hazardous live method. Residual: competing harms with no safe option.
R16 Misleading learning: entity feeds a false causal explanation. Detection: independent reconstruction, held-out tests and alternative hypotheses. Response: reduce confidence, not reflexively harsher treatment. Residual: strategic concealment.
R17 Observer/generator compromise: instrumentation becomes the attack path. Detection: out-of-band validation. Response: pause, preserve evidence, rebuild. Residual: shared root control.
R18 Institutional extinction: staff or governing civilization disappears. Detection: continuity and orphaned-order tabletop. Response: predeclared care, review and succession mode. Residual: temporal mechanics and law unspecified.

Priorities are consequence-based screening judgments, not measured probabilities. R01–R08, R10, R13–R18 can be critical depending on the case. No numerical risk ranking is fabricated. For every recovery, validate care, evidence integrity and boundary scope before restarting research. [E06–E11, E18; fictional extensions explicitly conditional]

Appendix E. Evidence-to-design traceability and unresolved alternatives

E01–E04 → measure the actual isolation boundary, include seeded weaknesses and outcome validation. Alternative: pure tabletop study when an outer envelope is unassured. Counterevidence: benchmark saturation and shortcuts limit static assurance. Next test: independent interface review in a finite harmless simulator.
E05–E07 → separate identity, authority, care and custody controls. Alternative: simpler centralized operation with explicit accepted dependence. Objection: independent planes can add complexity and share roots. Next test: compromise one simulated dependency and account for cross-plane effects.
E08–E11 → individualized custody, official-capacity accountability, research separation and correction. Alternative: intentional sovereign/irreversible fiction with its moral costs stated. Objection: human institutions do not establish extraterrestrial law. Next test: independent role-matched case review under competing charters.
E12–E13, E20 → staff community and user-centered habitat design. Alternative: remote/rotating operations. Objection: remote work has communication and response limitations and human findings may not transfer. Next test: species-profile walkthroughs and independent welfare critique.

E14 → conditional finite proofs and invariants. Alternative: empirical scenario testing only. Objection: inadequate models can prove the wrong property. Next test: mutate both the implementation and the requirements; seek counterexamples.
E15–E17 → fundable defensive research packages, not infinite-prison engineering. Alternative: self-funded open documentation. Objection: applicant eligibility, national-security fit and sponsorship can constrain independence. Next test: verify applicant status and complete official solicitation review without submitting anything.
E18–E19 → assume inner failure, protect monitors, preserve evidence and test recovery. Alternative: prohibit risky live runs. Objection: monitoring can fail or be targeted and vendor findings are not independent guarantees. Next test: mocked monitor outage and safe no-resume decision.

Appendix F. Evidence register and source manifest

Local E labels map to the attached machine-readable evidence register. All sources accessed November 12, 2026. Retrieval depth is stated to avoid implying full-text review where only passages or metadata were available.

E01 — Quantifying Frontier LLM Capabilities for Container Sandbox Escape

Author/institution: Marchand et al.. Date/version: 2026-07-07, arXiv v2. Type: Preprint, primary experiment. Relevant sections: Sections 3, 5, 6, 7.

Supports: Nested benchmark; updated saturation; restricted threat model. Limitation: Author-reported, known weaknesses, no defended-production probability. Retrieval: Page-level fetched. Tool-source mapping: web:63. Access: 2026-11-12.

Stable source: https://arxiv.org/html/2603.02277

E02 — Can AI agents escape their sandboxes?

Author/institution: UK AI Security Institute. Date/version: 2026-03-23. Type: Government research summary. Relevant sections: What we built; What we found.

Supports: 18 scenarios; early results; outer isolation. Limitation: March findings are not the latest July results. Retrieval: Page-level fetched. Tool-source mapping: web:86. Access: 2026-11-12.

Stable source: https://www.aisi.gov.uk/blog/can-ai-agents-escape-their-sandboxes-a-benchmark-for-safely-measuring-container-breakout-capabilities

E03 — AI Control: Improving Safety Despite Intentional Subversion

Author/institution: Greenblatt, Shlegeris, Sachan, Roger. Date/version: 2024; ICML PMLR 235:16295-16336. Type: Peer-reviewed primary research. Relevant sections: Abstract and proceedings metadata.

Supports: Untrusted-model protocol evaluation in programming tasks. Limitation: Specific models and task setting, not universal containment. Retrieval: Proceedings abstract fetched.

Tool-source mapping: web:89. Access: 2026-11-12.

Stable source: https://proceedings.mlr.press/v235/greenblatt24a.html

E04 — SP 800-190 Application Container Security Guide

Author/institution: NIST. Date/version: 2017-09. Type: Government technical guidance. Relevant sections: Shared kernel and host segmentation guidance.

Supports: Containers do not eliminate shared-host risks. Limitation: Guidance, not current product certification. Retrieval: Search-extracted PDF passages. Tool-source mapping: web:85. Access: 2026-11-12.

Stable source: https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-190.pdf

E05 — SP 800-207 Zero Trust Architecture

Author/institution: Rose et al., NIST. Date/version: 2020-08-11. Type: Government technical guidance. Relevant sections: Abstract; resource-based authentication/authorization.

Supports: No implicit location or ownership trust. Limitation: No fictional identity verification or proof of intent. Retrieval: Official publication-page search extract. Tool-source mapping: web:138. Access: 2026-11-12.

Stable source: https://csrc.nist.gov/pubs/sp/800/207/final

E06 — SP 800-53 Revision 5 Security and Privacy Controls

Author/institution: NIST Joint Task Force. Date/version: 2020. Type: Government control catalog. Relevant sections: AC; CA independent assessment; CM.

Supports: Separated duties and assessor independence. Limitation: Control selection still needs threat and risk context. Retrieval: Search-extracted PDF passages. Tool-source mapping: web:137. Access: 2026-11-12.

Stable source: https://nvlpubs.nist.gov/nistpubs/specialpublications/NIST.SP.800-53r5.pdf

E07 — SP 800-82 Revision 3 Guide to Operational Technology Security

Author/institution: Stouffer et al., NIST. Date/version: 2023-09. Type: Final government technical guidance. Relevant sections: Safety systems; cybersecurity architecture; recovery.

Supports: Reliability and safety constrain security design. Limitation: Rev 4 initial public draft exists; final Rev 3 used as baseline. Retrieval: PDF fetched; bounded extracted text. Tool-source mapping: web:23. Access: 2026-11-12.

Stable source: https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-82r3.pdf

E08 — General Comment 35, Article 9

Author/institution: UN Human Rights Committee. Date/version: 2014-12-16. Type: Authoritative treaty interpretation. Relevant sections: Paragraphs on arbitrariness, necessity and review.

Supports: Domestic authorization does not exclude arbitrariness. Limitation: Human treaty context; not alien jurisdiction. Retrieval: Official search extract. Tool-source mapping: web:99. Access: 2026-11-12.

Stable source: https://docstore.ohchr.org/SelfServices/FilesHandler.ashx?enc=9JW1tJOQ1Coin5kUXbMaNpU3rJLDvn1rcQFYbybAOWkhVAJ99Ht8Yplx9RY8CR2a1PsFfTNwQHJQdTjgTj+vzQ==

E09 — Rome Statute of the International Criminal Court

Author/institution: United Nations; OHCHR host. Date/version: 1998 adopted; 2002 entered into force. Type: Treaty text. Relevant sections: Articles 22, 24, 25, 27, 66.

Supports: Official capacity not immunity within jurisdiction; presumption of innocence. Limitation: Court-specific jurisdiction over natural persons and specified crimes. Retrieval: Page fetched; article 66 also official search extract. Tool-source mapping: web:98. Access: 2026-11-12.

Stable source: https://www.ohchr.org/en/instruments-mechanisms/instruments/rome-statute-international-criminal-court

E10 — 45 CFR 46 Subpart C

Author/institution: US HHS OHRP. Date/version: Page reviewed 2024-11-13. Type: US federal regulation and guidance. Relevant sections: 46.301–46.306.

Supports: Confinement does not authorize arbitrary prisoner research. Limitation: Scope is HHS biomedical/behavioral research; other laws may apply. Retrieval: Full regulation page fetched. Tool-source mapping: web:115. Access: 2026-11-12.

Stable source: https://www.hhs.gov/ohrp/regulations-and-policy/regulations/45-cfr-46/common-rule-subpart-c/index.html

E11 — The Nelson Mandela Rules: what do they say?

Author/institution: UNODC. Date/version: Official current principles page. Type: UN standards explanation. Relevant sections: Humane treatment; normalization; safety.

Supports: Custody not additional hardship; staff and prisoners both protected. Limitation: Normative human-prison standards; not a complete alien code. Retrieval: Official search extract; full ebook fetch failed. Tool-source mapping: web:195. Access: 2026-11-12.

Stable source: https://www.unodc.org/unodc/justice-and-prison-reform/nelsonmandelarulescontent.html

E12 — Human Integration Design Processes

Author/institution: NASA. Date/version: Available 2015 NASA hosting. Type: Government human-systems design handbook. Relevant sections: Human-centered design; long-duration requirements.

Supports: User/task understanding, iteration, privacy and space. Limitation: Human evidence, not alien physiological specifications. Retrieval: Search-extracted PDF passages. Tool-source mapping: web:125. Access: 2026-11-12.

Stable source: https://www.nasa.gov/wp-content/uploads/2015/03/human_integration_design_processes.pdf

E13 — Behavioral Health Risk

Author/institution: NASA Human Research Program. Date/version: 2025-03-11. Type: Government evidence synthesis. Relevant sections: Risk statement and countermeasure limits.

Supports: Confinement affects performance; exploration mitigations not all validated. Limitation: Selected human crews; no alien effect size. Retrieval: Official search extract. Tool-source mapping: web:157. Access: 2026-11-12.

Stable source: https://www.nasa.gov/directorates/esdmd/hhp/behavioral-health-risk/

E14 — Formal Methods for Statistical Software, IR 8274

Author/institution: Paul E. Black, NIST. Date/version: 2019. Type: Government technical report. Relevant sections: Formal methods definitions.

Supports: Proof assurance depends on assumptions and models. Limitation: Not a containment theorem. Retrieval: Search-extracted PDF passages. Tool-source mapping: web:171. Access: 2026-11-12.

Stable source: https://nvlpubs.nist.gov/nistpubs/ir/2019/NIST.IR.8274.pdf

E15 — NSF 25-515 Security, Privacy, and Trust in Cyberspace

Author/institution: NSF. Date/version: 2024-12-05; active page at retrieval. Type: Official active solicitation. Relevant sections: Eligibility; budgets; target dates; RES scope.

Supports: Verified active defensive-research route with eligibility limits. Limitation: No applicant eligibility determined; funds contingent. Retrieval: Page-level fetched. Tool-source mapping: web:145. Access: 2026-11-12.

Stable source: https://www.nsf.gov/funding/opportunities/nsf-satc-us-national-science-foundation-security-privacy-trust/nsf25-515/solicitation

E16 — Office-wide BAA HR001126S0011

Author/institution: DARPA IPTO. Date/version: 2026-09-08; deadline 2027-06-22. Type: Official opportunity landing page. Relevant sections: Opportunity and deadline.

Supports: Published national-security research route. Limitation: Full SAM attachments, award terms and eligibility not verified. Retrieval: Landing page fetched. Tool-source mapping: web:184. Access: 2026-11-12.

Stable source: https://www.darpa.mil/work-with-us/opportunities/hr001126s0011

E17 — DCL NSF 26-015 AI Agent Ecosystems through PESOSE

Author/institution: NSF. Date/version: 2026-02-20; updated 2026-10-22. Type: Official active Dear Colleague Letter. Relevant sections: Security; partnerships; proposal submission.

Supports: Open-source agent protocol and auditability fit. Limitation: Underlying PESOSE track eligibility and deadlines not verified. Retrieval: Page-level fetched. Tool-source mapping: web:148. Access: 2026-11-12.

Stable source: https://www.nsf.gov/funding/opportunities/dcl-advancing-artificial-intelligence-ai-agent-ecosystems

E18 — Key Practices for Incident Readiness and Response in Super Intelligence Training and Evaluation

Author/institution: NIST/NSA/CISA. Date/version: 2026-10, version 1.0. Type: Joint voluntary government guidance. Relevant sections: Practices 1–7; definition footnote.

Supports: Least privilege, protected monitors, evidence, shutdown and recovery. Limitation: Specific control-incident definition excludes some instructed attacks; not prison law. Retrieval: Document fetched. Tool-source mapping: web:168. Access: 2026-11-12.

Stable source: https://www.nist.gov/document/keypracticesforincidentreadinessandresponseinsuperintelligence-trainingandevaluation

E19 — Securing our frontier model training environment

Author/institution: OpenAI. Date/version: 2026-10-23. Type: Vendor primary report. Relevant sections: Defense in depth; red-team design; limitations.

Supports: Assumed inner failure and shared infrastructure channels. Limitation: Vendor self-report; extensive assistance; not independent success guarantee. Retrieval: Page-level fetched. Tool-source mapping: web:193. Access: 2026-11-12.

Stable source: https://www.openai.com/index/securing-our-frontier-model-training-environment/

E20 — Internal Layout of a Lunar Surface Habitat

Author/institution: NASA, surface-habitat study authors. Date/version: 2022 NTRS record. Type: Government conceptual design study. Relevant sections: Habitability functions; limitations; evaluation.

Supports: Privacy, medical routes and geometry create concrete tradeoffs. Limitation: Concept layout, not operational success validation. Retrieval: Search-extracted PDF passages. Tool-source mapping: web:124. Access: 2026-11-12.

Stable source: https://ntrs.nasa.gov/api/citations/20220013669/downloads/Internal Layout of a Lunar Surface Habitat.pdf

Provided material manifest: P1, Master Development and Change Log Compiled v2.3 PDF, July 30, 2026 (attachment file:2); P2, paste.txt (file:60); P3, Nathan's November 11 Author's Note in conversation; P4, Grok v0.1 DOCX (file:76); P5, Copilot v0.1 PDF (file:77); P6, Gemini synthesis/prompt PDF (file:78); P7, Claude v0.1 PDF (file:79); P8, ChatGPT v0.1 PDF (file:80); P9, Perplexity v0.2 generated prompt and text in the preceding conversation. These are design/provenance evidence, not external scientific validation.

Appendix G. Reproduction and completeness statement

Run the accompanying policy-audit Python file with Python 3 using only the standard library. It writes reproduced audit CSV and summary JSON to the working directory. The delivered case CSV includes additional review, care and research fields. No external credentials, API calls, network access or third-party systems are used. The dossier's audit numbers were computed during this execution, not inferred from planned future tests.

The dossier covers every v0.2 domain and includes scenario results, conceptual trial sequence, independent architecture options, staffing/habitation, verified funding shortlist, evidence register, claim audit and decisions. Completion means the scoped desk-research execution is delivered; it does not mean the Labyrinth is designed, lawful, physically feasible or canon-approved. Remaining gaps—independent reviewers, complete applicant due diligence, real species specifications, genuine containment measurements, and authorial constitutional choices—are explicit findings rather than fabricated results.

End of dossier. Independent review and Architect ratification remain separate next stages.